

DAY 1 : MYSQL

What is data?

Raw facts or figures without context

Eg: 200,praveen,vizag

What is field?

Small unit of data in database(column/attribute)

Eg:

Student_name,Student_age

What is record?

Collection of related fields(row/tuple)

Eg: (100,'Sunny',20000)

What is Database?

Organized collection of related records stored together

Eg: An student database containing all student records.

Client-server-architecture

Client: The role of the client in client-server architecture is to send request to the server and to get response from the server.

Server: Server is a software the role of the server in client-server arch is to take request from the client. If the request is valid Request it will generate some response and we can handover that response to client. If the client request is not a valid request it Will raise some error status codes.

Protocol: It has a set of rules and regulations. The role of the protocol in client- server architecture is to send request from client to server

Identifiers

Identifiers are name used to identify database object such as, Database names, table names etc..

Rules of Identifiers:

- characters to declare identifiers a-z or A-Z or 0-9,_,#.
- Identifiers should not start with digit.
- Spaces are not considered to declare identifiers.
- Reserved words not considered to declare identifiers.
- max length to declare identifiers are 15 characters.

SQL(Structured Query Language)

Structured Query Language(SQL) is a Standardized programming language specially designed for managing and manipulating relational databases. It provides a declarative syntax that allows users to define what data they want retrieve, insert, update or delete without specifying exactly how database should execute these operations.

SQL Commands

----- DDL: Data Definition Language

Define and modifies the structure of database objects (tables, schemas, indexes)

Create: Creates a new database objects such as tables, database, views, indexes or triggers... etc

How to create database/schema?

Syntax: CREATE DATABASE database-name;
Or
CREATE SCHEMA schema-name;

How to create table?

Syntax:

CREATE TABLE table-name(
Column-name datatype(size) constraint,
Column-name datatype(size) constraint,
Column-name datatype(size) constraint,
Column-name datatype(size) constraint);

Drop: Permanently deletes an existing database objects (table, database, index, views) along with its data and structure.

How to drop database/schema?

Syntax:

Drop database database-name;
Or
Drop SCHEMA schema-name;

How to drop a table?

Syntax:

Drop table table-name;

DML: Data Manipulation Language

Manages data stored within database objects (insert, update, delete)

DCL: Data Control Language

Manages permissions and accesses control on database objects.

TCL: Transaction Control Language

Manages transaction to maintain data integrity.

DQL: Data Query Language

Used to fetch/retrieve data from the database.